

Suraj Pratap Singh

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SUMMARY

Flutter and full-stack engineer building Synapse, an intelligent action router and plugin platform that turns screenshots, links, text, and audio into automated workflows. Experienced across mobile architecture, plugin SDK design, backend APIs, AI-assisted capture pipelines, and local persistence using Dart, TypeScript, NestJS, Supabase, and PostgreSQL.

SKILLS

Languages: Dart, TypeScript, Python, Java, C++, SQL, JavaScript

Mobile / Frontend: Flutter, React, Next.js, Tailwind CSS, Bloc, go_router, share-sheet integrations

Backend / Platform: NestJS, Node.js, Spring Boot, Flask, Supabase, PostgreSQL, Redis, OAuth

AI / Automation: OCR, ASR, LLM orchestration, plugin execution, RAG, LangChain, PyTorch, TensorFlow, YOLOv5

Tools: Git, Docker, Jira, Bitbucket, Confluence, DBeaver

SELECTED PROJECT

Synapse - Intelligent Action Router and Plugin Platform

*Flutter, Dart, TypeScript, NestJS,
Supabase, PostgreSQL, OCR,
ASR, LLMs*

- Building a cross-platform Flutter host that routes screenshots, links, text, and audio into executable actions through OCR, speech transcription, LLM intent detection, and plugin dispatch.
- Designed the plugin SDK and .synx packaging flow with host-managed auth, storage, UI, file upload, and fetch primitives for third-party plugin authors.
- Implemented a marketplace backend for plugin submission, review, versioning, signed storage, and MCP registry publishing.

EXPERIENCE

Sapiens Technologies

Jan 2024 - Jul 2025

Associate Software Developer

Bengaluru, India

- Built and enhanced insurance accounting features for the IDITSUITE platform as part of a cross-functional team.
- Refactored legacy Java (EJB, Struts) modules to Spring Boot and React, improving scalability, maintainability, and user experience.
- Optimized SQL queries and backend logic, achieving 5x faster queries and an 80% performance improvement.
- Collaborated with QA using Jira, Bitbucket, and Confluence to streamline testing and release cycles.
- Designed and documented complex business logic across multi-layered systems, improving implementation clarity and peer reviews.

Smart Bridge

May 2023 - Jul 2023

ML (Python, AI/ML)

Bengaluru, India

- Engineered a Virtual Eye-Lifeguard system using YOLOv5 for real-time drowning detection.
- Reached 70% incident detection accuracy through data annotation, model training, and performance evaluation.

ADDITIONAL PROJECTS

Drip YouTube Music Client

*Dart, Flutter, Dart FFI, C++,
Python, FluentUI, Flask, Hive*

- Developed a cross-platform YouTube Music client using Flutter and Dart with a modern desktop-first interface.
- Reverse-engineered the YouTube Music API in Python and Dart, integrated LibMpv via Dart FFI, and published the project on GitHub with 150+ stars.

Nimbus

*Python, Node.js, PostgreSQL,
Llama 3.2B, Redis, Docker,
LangChain*

- Built an end-to-end app that scrapes RSS feeds and summarizes articles into concise 100-word summaries.
- Implemented a Python scraper plus a queue-driven Node.js and Redis backend, then deployed a fine-tuned Llama 3.2B model in Docker.

ACHIEVEMENTS

- Two-time finalist at Neo-Code-Athon (iamneo) DSA hackathon, ranking 3rd among top 10% performers.
- Founder of Coffee Coders Collective (CCC), mentoring 30+ students in Flutter and full-stack development.

EDUCATION

Master of Science - Information Technology (AI)

Sep 2025 - Present

UNSW Sydney

Sydney, NSW | WAM: 80 (Distinction)

Bachelor of Technology

Aug 2020 - Jul 2024

Vellore Institute of Technology

Bhopal, MP (India) | CGPA: 8.55 / 10